Hadronic and Coulomb losses for streaming cosmic rays, a 20 kpc tall box and compressed output: change log

Base commit **fc81616ed** · Date **1 November 2026** · Prepared for **Vinod V. Pisharody**

This commit adds the first energy sink of the two-moment (streaming) cosmic rays: hadronic and Coulomb collisions with the gas. It also adds the diagnostics and the setup used for the run D10: MHD + streaming CR with losses, no field-SN driving, on a 1 × 1 × 20 kpc box. The motivation came from the 4 kpc runs C1, D and Dnd. There the CR energy grew without limit and came to dominate the energy budget. Their CR halo was also larger than the box, so part of it was cut off by the boundary. D10 ran 0 → 54.1 Myr on this source and was stopped on request. Section B gives the validation, section C what is still open and section D the state of the run.

6
records in 7 files: 4 source, 1 parameter file, 1 build log (and this PDF)

0.4%
agreement between the drop in E_{CR} and the integrated loss luminosity (losses on vs off)

54.1 Myr
of the 20 kpc production run D10 (8 194 steps), no NaN, no multigrid warning

0
HDF5 error messages with compression after the fix (3 010 before)

PHYSICS new physics, off by default · BUG FIX wrong behaviour · DIAGNOSTIC output only, no effect on results · CONFIG parameter files, logs · OPEN found, not changed here

Contents

1. CR loss parameters and rates · initstreamingcr.F90
2. CR losses in the source step · streamingcr_source.F90
3. Loss and γ-ray luminosity in the tsl · tallbox_sf/initproblem.F90
4. Mass outflow at four heights · tallbox_sf/initproblem.F90
5. No compression for empty datasets · common_hdf5.F90
6. The D10 setup and its build log · problem.par.D10, build_D10_20pc.log

Then: **B** validation · **C** open issues · **D** state of the run.

A. Overview: before, after, motivation

#	BEFORE	AFTER	MOTIVATION
1	No collisional losses for streaming CRs	Namelist switch `cr_losses` (default off); coefficients ζ_h, ζ_C, μ_e; rate and heating and γ shares computed once	CR energy had no sink and came to dominate the energy budget
2	—	E_c and F_c decay as $\exp(-(\zeta_h+\zeta_C)\, n_e\, \Delta t)$; Coulomb losses and ⅙ of the hadronic ones heat the gas	Exact and unconditionally stable for any sub-step; energy goes where the physics sends it
3	—	tsl columns `L_cr_loss`, `L_gamma`	CR energy budget and γ-ray luminosity of the run
4	Outflow at |z| = 0.5, 1 kpc and the boundary	Also at 2 and 5 kpc (`Mdot_z3`, `Mdot_z4`)	With a ±10 kpc box, 1 kpc and the boundary are too far apart to tell a fountain from a wind

5	Chunked, compressed dataset even when one dimension is 0	No chunking or compression for empty datasets	HDF5 rejects zero-sized chunks and printed 3 010 error lines per run
6	Only the ±2 kpc setups	`problem.par.D10`: ±10 kpc, losses on, undriven, CG coarse solver, compressed output	The CR halo outgrew the ±2 kpc box; the 1:20 box defeated the multigrid; disk space

Records

1 CR loss parameters and rates

PHYSICS

src/fluids/cosmicrays_streaming/initstreamingcr.F90 · lines 76–79, 106–109, 133, 161–164, 203–214, 256–267, 293–302

AFTER (new namelist parameters of STREAMING_CR, broadcast as rbuff(10:12) and lbuff(8), nl = 8)

```
   logical :: cr_losses   !< hadronic + Coulomb losses of the CR energy (and flux) on the gas
   real    :: zeta_hadr   !< hadronic loss coefficient [cm^3/s]: dEc/dt = -zeta_hadr n_e Ec (Pfrommer et
al. 2017)
   real    :: zeta_coul   !< Coulomb loss coefficient [cm^3/s]: dEc/dt = -zeta_coul n_e Ec
   real    :: mu_e_loss   !< mass per electron in units of m_p, n_e = rho / (mu_e_loss m_p)
   ...
      cr_losses = .false.  ;  zeta_hadr = 7.44e-16  ;  zeta_coul = 2.78e-16  ;  mu_e_loss = 1.17
   ...
      if (cr_losses) then
         cr_loss_rate       = (zeta_hadr + zeta_coul) * cm**3 / sek / (mu_e_loss * mp)
         cr_loss_heat_frac  = (zeta_coul + zeta_hadr / 6.0) / (zeta_hadr + zeta_coul)
         cr_loss_gamma_frac = (zeta_hadr / 3.0) / (zeta_hadr + zeta_coul)
         ! printinfo: "CR losses: rate = 1.115E+00 * rho per code time, heating share .393, gamma-ray
share .243"
```

WHY

In the ±2 kpc runs, CR energy was only removed by the boundary and by its small exchange with the gas. In the undriven run Dnd it reached 78×10^{51} erg at 54 Myr, more than the thermal and kinetic energy together (70×10^{51} erg). Collisions with gas nuclei and electrons are the main CR energy sink in the ISM. For a CR population with the usual spectrum, Pfrommer et al. (2017, MNRAS 465, 4500) give the loss rates per unit CR energy density: $\Lambda = \zeta\, n_e\, \varepsilon_c$, with $\zeta_h = 7.44 \times 10^{-16}$ and $\zeta_C = 2.78 \times 10^{-16}$ cm^3 s^{-1}.

- **Where the energy goes.** Coulomb losses heat the gas entirely. Of the hadronic losses, ⅙ goes to secondary electrons and positrons, which heat the gas; ⅓ leaves as γ-rays from π^0 decay; the rest leaves as neutrinos. So 39.3% of the lost energy heats the gas and 24.3% is γ-rays.
- **Electron density.** $n_e = \rho / (\mu_e\, m_p)$ with $\mu_e = 1/(X + Y/2) \approx 1.17$ for fully ionised gas with $X = 0.7$.
- **Rate in code units.** The rate is 1.115 ρ per Myr, with ρ in $M_\odot$ pc^{-3}. That gives a loss time of 31 Myr at $n_e = 1$ cm^{-3}.
- **Defaults.** The switch is off by default, and the loss code is only reached when it is on, so existing setups are unaffected by construction (not tested bit for bit, section B).

2 CR losses in the source step

PHYSICS

src/fluids/cosmicrays_streaming/streamingcr_source.F90 · lines 44–51, 73, 235–253

AFTER (after the floor, the clip, the CR–gas exchange and the last-resort floor; before Ec, Fc are written back)

```
            ! hadronic + Coulomb losses, dEc/dt = -(zeta_hadr + zeta_coul) n_e Ec, integrated exactly
over the sub-step; Fc decays
            ! at the same rate. Coulomb losses and 1/6 of the hadronic ones (secondary e+-) heat the
gas, the rest leaves as
            ! gamma rays and neutrinos.
            if (cr_losses) then
               floss = exp(-cr_loss_rate * max(smalld, cg%w(fldi)%arr(iarr_all_dn(1),i,j,k)) *
rk_coef(istep) * dt_scr)
               de_l  = newec * (1.0 - floss)
               if (use_smallescr) de_l = min(de_l, max(newec - smallescr, 0.0))
               de_l  = max(de_l, 0.0)
               newec = newec - de_l
               newf1 = newf1 * floss
```

```
               newf2 = newf2 * floss
               newf3 = newf3 * floss
#ifndef ISO
               if (.not. disable_feedback) &
                    & cg%w(uhi)%arr(iarr_all_en(1), i, j, k) = cg%w(uhi)%arr(iarr_all_en(1), i, j, k) +
cr_loss_heat_frac * de_l
#endif /* !ISO */
               if (istep == last_stage(integration_order) .and. i >= cg%is .and. i <= cg%ie .and. j >=
cg%js .and. j <= cg%je &
                    & .and. k >= cg%ks .and. k <= cg%ke) scr_eloss = scr_eloss + de_l * cg%dvol
            endif
```

WHY

- **Integration.** The loss is linear in E_c, so it is integrated exactly over each CR sub-step and RK stage (rk_coef × dt_scr, the same time step as the rest of this source routine). The factor exp(−rate Δt) is between 0 and 1 for any Δt, which matters because in dense gas (n ≈ 100 cm^{-3}, loss time 0.3 Myr) an explicit update would need its own time-step limit.
- **Flux.** F_c decays by the same factor. That keeps $|F_c|/E_c$, so the streaming direction and the causality limit set just before are unchanged.
- **Placement.** The block comes after the CR–gas exchange of 9e804b91d, so that exchange still pays exactly for the transport change of E_c and F_c. The loss is a separate sink acting on the final E_c.
- **Floor and heating.** The loss cannot push E_c below `smallescr`. The heating share goes into the total gas energy of the stage output (`uhi`). It is skipped with `disable_feedback` and under ISO, where there is no energy equation.
- **Bookkeeping.** The lost energy is counted only at the last RK stage, which carries the actual update, and only in interior cells, so ghost cells are not counted twice.

3 Loss and γ-ray luminosity in the tsl

DIAGNOSTIC

`problems/tallbox_sf/initproblem.F90 · lines 60, 701–703, 721–724, 737, 795–800, 821–822`

AFTER

```
      qs = [real([scr_nclip_src, scr_nclip_face, scr_nfallback, scr_npp, scr_nfloor, scr_nexch_cap]),
scr_eloss]
      call piernik_MPI_Allreduce(qs, pSUM)
      ... ; scr_eloss = 0.0
      lloss = 0.0                                          ! CR energy lost per unit time since the
previous dump
      if (t_tsl_last >= 0.0 .and. t > t_tsl_last) lloss = qs(7) / (t - t_tsl_last)
      t_tsl_last = t
      ...
         call pop_vector(user_vars, [cred, qs(1:6)])
         call pop_vector(user_vars, [lloss, cr_loss_gamma_frac * lloss])     ! tsl: L_cr_loss, L_gamma
```

WHY

Each tsl row records the mean loss luminosity since the previous row. ∫L dt then closes the CR budget: injected = E_{CR} + lost + the rest, where the rest is what leaves the box plus what the transport exchanges with the gas. $L_\gamma = 0.243\ L$ can be compared with the γ-ray–SFR relation. The safeguard counters keep their order. Because `t_tsl_last` is not stored in restart files, the first row after a start or restart reads 0. That row repeats the time of the last row before the restart, so nothing is lost from the integral.

4 Mass outflow at four heights

DIAGNOSTIC

`problems/tallbox_sf/initproblem.F90 · lines 51, 120, 161, 195, 675–677, 711, 731–733, 743–744, 770–772`

BEFORE → AFTER

```
   real, dimension(2) :: z_flux ;  z_flux = [500.0, 1000.0]
   ->  real, dimension(4) :: z_flux ;  z_flux = [500.0, 1000.0, 2000.0, 5000.0]     (rbuff(22:25))
   do l = 1, 2 ... q(5+l)   ->   do l = 1, size(z_flux) ... q(merge(5+l, 10+l, l <= 2))      ! nq 12 ->
14
   new tsl columns Mdot_z3, Mdot_z4, appended after Mgas
```

WHY

In a ±10 kpc box the old heights, 0.5 and 1 kpc, and the boundary are too far apart to follow the outflow as it moves up. In D10 the net mass loading falls from 15.5 at 0.5 kpc to 3.9 (1 kpc), 1.3 (2 kpc), 0.5 (5 kpc) and 0 at 10 kpc. The new heights also show the front: it passed 2 kpc at about 38 Myr and 5 kpc at about 48 Myr. The new columns come after the existing ones, so scripts that read columns by position still work. Parameter files that give only two values keep the new defaults for the other two. In a ±2 kpc box the third height falls on the boundary layer and the fourth lies outside, where it reads 0.

5 No compression for empty datasets

BUG FIX

`src/IO/hdf5/common_hdf5.F90 · line 961`

BEFORE

```
      if (enable_compression .and. Z_avail) then
```

AFTER

```
      if (enable_compression .and. Z_avail .and. all(ddims > 0)) then   ! HDF5 refuses zero-sized chunks
(e.g. a block without particles)
```

WHY

With `enable_compression`, every dataset is created chunked, with the chunk set to the full dataset size. A grid block without star particles has particle datasets of length 0, which HDF5 refuses as a chunk size. At t = 0 that is every block. HDF5 then printed its error stack: 3 010 HDF5-DIAG lines in D10's run.log, at every snapshot and restart. The datasets were still written and readable, and a restart from such a file was checked to read back correctly. Empty datasets now skip chunking and compression, since there is nothing to compress. D10 ran with the binary from before this fix (section D).

6 The D10 setup and its build log

CONFIG

`problems/tallbox_sf/problem.par.D10 (new) · build_D10_20pc.log (new)`

The setup is `problem.par.D` with the changes below. Build: `./setup tallbox_sf -o D10_20pc -d H5PFC_EATS_1ST,STREAM_CR -p problem.par.D10`; run with `-n '&STREAMING_CR cred_min = 1.0e4 /'`, as D was.

PARAMETER	D	D10	WHY
`n_d`, `zmin/zmax`	48,48,192; ±2 kpc	48,48,960; ±10 kpc	Same 20.8 pc cells. At 50 Myr, 32% of D10's CR energy lies above |z| = 2 kpc (H_{CR} ≈ 1.8 kpc), and Dnd's boundary drained 41% of the injected CR energy
`cr_losses`	off	on	Records 1–2
`t_drive_end`, `t_taper`	40, 10	−1, 0	No field-SN driving, as in Dnd, so that D10 differs from Dnd only in box height and CR losses

`use_CG` (MULTIGRID_GRAVITY)	—	.true.	With a 1:20 box, relaxation on the coarsest level no longer converged. The V-cycles gave up with "Poor convergence … norm_tol missed by 0.2E+06" and self-gravity failed. A conjugate-gradient solve there fixes it: 0 warnings in 54 Myr, at no measurable cost per step
`enable_compression`, `gzip_level`	—	.true., 4	The box has 5× more cells. Snapshots at t = 0 shrink from 88.6 to 21.7 MB and are 61–73 MB later; restart files are 0.70 GB
`vars`	24 fields	11 fields	Disk: dens, vel, pres, temp, B, escr, zfscr are what the analysis uses
`dt_hdf`, `dt_res`, `tend`, `run_id`	5, 50, 500, D_s	10, 25, 200, D10	Disk and wall time (about 1–1.4 Myr/h on 6 ranks)
`z_flux`	0.5, 1 kpc	0.5, 1, 2, 5 kpc	Record 4

`build_D10_20pc.log` is the compiler log of the binary D10 ran with, kept as for C1 and D (commit fc81616ed). That build predates record 5.

B. Validation

Loss test. Run: the D10 setup on a ±2 kpc box (48 × 48 × 192), with losses on and off, from 0 to 2.14 Myr on 6 ranks. Energies are in code units.

CHECK	EXPECTED	MEASURED	READING
Loss luminosity, first tsl interval (0–0.33 Myr)	5.041×10^4	5.042×10^4	Expected value: 1.115 Σ ρ E_c dV from the snapshots, 5.090 at t = 0 and 4.991 at 0.33 Myr, averaged. Agreement 0.02%
ΔE_{CR}(on) − ΔE_{CR}(off), 2.14 Myr	-1.0253×10^5 ($-\int L\,dt$)	-1.0212×10^5	The extra CR energy lost with losses on equals the integrated luminosity to 0.4%
L_γ / L	0.2427	0.2427	$(\zeta_h/3)/(\zeta_h+\zeta_C)$
E_{int}(on) − E_{int}(off)	≤ 4.03×10^4 (0.393 $\int L\,dt$)	2.10×10^4	The gas keeps 52% of the heat. Losses are largest in dense gas, where the cooling time is short, so the rest is radiated. Not a conservation test: cooling and heating cannot be separated here

Binary built from this commit's tree (`obj_commit_chk`, compiled without errors in 39 s):

TEST	RESULT
Loss test above, losses on, first 0.33 Myr	**bit-identical** to the D10 binary: E_{CR} and L_{loss} = $5.04204617018 \times 10^4$ in both
Compressed snapshots and restart (record 5)	**0 HDF5-DIAG lines** (D10's binary: 3 010 lines in run.log)
Restart from the compressed restart file at 0.33 Myr, run to 0.75 Myr	**bit-identical** to the uninterrupted run at t = 0.651 Myr (E_{CR}, L_{loss})
Default `cr_losses = .false.` against the previous commit's binary	**Not tested.** The loss code is only reached with the switch on, and the switch is a new namelist entry, so setups that do not set it run unchanged. A bit-for-bit comparison was not run

Production run D10 (this source, apart from record 5):

QUANTITY	VALUE	READING
Run	0 → 54.107 Myr, 8 194 steps, 502 tsl rows	No NaN, no "Poor convergence" from the multigrid; stopped on request with a final restart
CR energy: in the box / lost / rest	81% / 14% / 5%	Of 148×10^{51} erg injected (10% of 1 484 SNe). In Dnd (±2 kpc, no losses) it was 59% / — / 41%, most of that 41% through the boundary
Loss time E_{CR}/L_{loss} at 54 Myr	≈ 220 Myr	CRs see an average $n_e \approx 0.14$ cm^{-3}, about ¼ of the midplane value
Energy added by the internal-energy floors	14×10^{51} erg	0.95% of the SN energy (0.5–0.6% in C1 and D)
Last-resort E_c floor (`nfloor`)	0	Never reached

CR exchange capped (`nexch_capped`) · faces with first-order fallback	3.3×10^4 · 2 616	Both below the D run's counts (4.9×10^5 · 1.3×10^4 over 102 Myr)
Alfvén-speed ceiling: mass added	72 $M_\odot$	7×10^{-6} of the gas mass

C. Open issues (not changed in this commit)

OPEN **1. The ±10 kpc boundary has not met the outflow yet.** At 54 Myr the outflow front was at about 5 kpc, moving at about 300 km s^{-1}, and the net flux through 10 kpc was a slight inflow (−16 $M_\odot$ Myr^{-1}). The gas and CR outflow boundaries at 10 kpc are therefore not yet tested under outflow. Whether this is a fountain or a wind is also undecided: the Ferrière potential keeps deepening with height, so reaching 10 kpc from the midplane needs about 290 km s^{-1}.

OPEN **2. The CR outflow boundaries still differ between the modules** (issue 2 of the previous change log): absorbing in `bnd_u` for one-moment CRs (C1) and zero-gradient in `bnd_scr` for two-moment CRs (D, D10). The losses exist only in the streaming module, so a C1-type run cannot yet be compared with losses on.

OPEN **3. Losses for one CR energy group.** ζ_h and ζ_C are fixed values for an assumed CR spectrum. The spectrum does not evolve, although Coulomb losses mainly remove low-energy CRs and hadronic losses high-energy ones. L_γ is the total π^0-decay luminosity, not the 0.1–100 GeV band of Fermi.

OPEN **4. Restart bookkeeping.** `t_tsl_last` is not saved, so the first L row after a restart reads 0 (record 3). The cumulative SF counters are still not saved either (issue 3 of the previous change log).

D. State of the run

RUN	STOPPED AT	RESTART FILES	NOTES
D10 (MHD + streaming CR + losses, ±10 kpc, undriven)	t = 54.107 Myr, step 8 194, 1 Nov 2026 12:04, on request	tallbox_sf_D10_0003.res (54.1 Myr), tallbox_sf_D10_0002.res (50.0 Myr)	Ran with the binary built before record 5. The binary of this commit is `obj_commit_chk/piernik`. 1.8 GB free on disk, not enough for the remaining 146 Myr at the current output rate